# Critical exponential metrics in higher dimensions

Euclidean topology and the subcritical limit

**Abstract**

Let $d \geq 3$ and let $h$ be a whole-space log-correlated Gaussian field. We prove that its critical exponential metric is finite everywhere and induces the Euclidean topology, with an almost sure logarithmic modulus on each compact set. We also prove that the median-normalized subcritical metrics, coupled through the same field, converge almost surely and locally uniformly to the critical metric through all real parameters. The topology proof constructs small random separators from coordinate cube boundaries and sums their costs by a Gaussian near-maximum occupation estimate. The endpoint proof first obtains convergence on deterministic hypersurfaces, using a comparison along geodesics, and then constructs separators simultaneously for the parameter family. We use the fixed-parameter construction and comparison methods of the public project manuscripts cited below; the estimates and modification needed at the critical endpoint are proved here.

## 1 Introduction

Let $d \geq 3$ and let $h$ be the whole-space log-correlated Gaussian field with logarithmic covariance coefficient one. We fix its additive constant by requiring its average on the Euclidean unit sphere to vanish. Thus, for smooth compactly supported zero-mass test functions,

$$\operatorname{Cov}(h(\varphi), h(\psi)) = \iint \varphi(x)\psi(y) \log \frac{1}{|x-y|}\, dx\, dy.$$

Write $q = \sqrt{2d}$ and define the critical Weyl parameter by

$$c = \zeta_c := \sup\{\zeta > 0 : Q_d(\zeta) > q\}.$$

The function $Q_d$ is continuous and nonincreasing, so $Q_d(c) = q$ [1, Section 1.2]. This definition specifies the endpoint of the subcritical regime even if $Q_d$ has a plateau at level $q$. The parameter $\zeta$ is the coefficient in Weyl scaling, and $\phi(\zeta) = \zeta Q_d(\zeta)$ is the similarity exponent. We consider the strong exponential metrics constructed in [1]. This construction applies whenever $Q_d(\zeta) > \sqrt{2(d-1)}$, and therefore includes the critical parameter.

Normalize each metric by the lower median of its unit distance:

$$\operatorname{med} D_h^{\zeta}(0, e_1) = 1. \tag{1.1}$$

All parameters are coupled through the same field. The first result identifies the topology at criticality and gives a quantitative upper modulus.

**Theorem A** (Critical topology)**.** Almost surely, $D_h^c$ is finite on $\mathbb{R}^d \times \mathbb{R}^d$ and induces the Euclidean topology. For every compact $K \subset \mathbb{R}^d$ and every $0 < \theta < c/(2\sqrt{2d})$, there is an almost surely finite $C_{K,\theta}$ such that

$$D_h^c(x,y) \le C_{K,\theta} \left( \log \frac{1}{|x-y|} \right)^{-\theta}, \qquad x, y \in K, \quad 0 < |x-y| \le \tfrac{1}{2}. \tag{1.2}$$

In particular, the critical metric has no singular points.

Lemma 8.1 constructs a field-measurable subcritical modification which is almost surely continuous in the locally uniform topology on a deterministic left neighborhood of $c$. Our second result extends this version to the critical endpoint. All limits below are through real parameters, on one event of probability one.

**Theorem B** (The subcritical limit)**.** The median-normalized subcritical metrics admit a same-field modification such that, almost surely,

$$\lim_{\zeta \uparrow c} \sup_{x,y \in K} |D_h^\zeta(x,y) - D_h^c(x,y)| = 0 \qquad \text{for every compact } K \subset \mathbb{R}^d. \tag{1.3}$$

Moreover, for every $0 < \theta < c/(2\sqrt{2d})$, there is a deterministic $\varepsilon_\theta > 0$ such that almost surely

$$\sup_{c - \varepsilon_\theta \le \zeta \le c} D_h^\zeta(x,y) \le C_{K,\theta} \left( \log \frac{1}{|x-y|} \right)^{-\theta} \tag{1.4}$$

for $x, y \in K$ with $0 < |x-y| \le 1/2$.

Theorem B concerns uniform convergence on compact sets in the two spatial variables. It asserts no rate in $c - \zeta$. A modification agrees with the prescribed metric almost surely at each deterministic parameter. The subcritical continuous modification is fixed before taking the endpoint limit.

### 1.1 The argument and its sources

At criticality the usual bulk Hölder exponent vanishes. The positive gap $q - \sqrt{2(d-1)}$ nevertheless leaves a Hölder estimate on deterministic coordinate hypersurfaces. We construct random separators from small cube boundaries and obtain a sharp enough tail for their metric diameters. A Gaussian maximum estimate and a Brownian-bridge occupation estimate make the costs summable uniformly over space. This uses the near-maximum mechanism of the planar critical-topology proof [3], with the higher-dimensional separator geometry proved here, and yields Theorem A.

For the endpoint, uniform marginal estimates first give convergence in probability. To control parameter oscillations, we compare metrics on fixed hypersurfaces. The critical comparison uses regularity along geodesics with endpoints on these surfaces; it does not require a positive bulk Hölder exponent. On strictly subcritical bands, explicit tracking of the existing comparison proof gives constants bounded by an exponential tower of fixed height. Parameter grids then give almost sure convergence on every rational coordinate complex. Their intrinsic diameters become bounded simultaneously over the parameter, so a single fine-field percolation construction yields the common spatial modulus. A finite spatial net completes the proof.

We use the following established conclusions of the project papers, with citations to their public versions. The assignment in [1, Definition 1.3 and Theorem 1.4] is a separating, Euclidean-lower-semicontinuous extended length metric; its actual intrinsic metrics are local, and it satisfies Weyl scaling and exact deterministic similarities. Finite metric paths are Euclidean continuous by [1, Proposition 11.3(T5)]. Intrinsic surface estimates for deterministic coordinate complexes follow from [1, Proposition 11.3(T2), Lemmas 12.6 and 12.9, and Proposition 13.1(a)]. The path-length Weyl identity and dependent continuous perturbations are covered by [1, Lemma 12.13 and Proposition 13.7].

For positive-parameter continuity we use the construction in [2, Sections 5.1, 5.3–5.4], with its inputs supplied here. The fixed-parameter characterization is [1, Definition 1.3 and Theorem 1.4]; Sections 5 and 7 provide the marginal and increment estimates. Lemma 8.1 uses parameter grids suited to those estimates and constructs the common version directly. Lemma 10.1 supplies median uniqueness. This application therefore does not depend on the older characterization or the separate small-parameter manuscript cited in [2].

### 1.2 Notation

Write $B_r(z) = z + (-r, r)^d$, $B_r = B_r(0)$ and $K_R = \overline{B}_R$. We use $|\cdot|$ for the Euclidean norm and $|\cdot|_\infty$ for the maximum norm. The notation $h_r(z)$ denotes the field average on the Euclidean sphere of radius $r$ centered at $z$. For a compact set $S$, let $\|F\|_S = \sup_{x,y\in S} |F(x,y)|$. The notation $D(\cdot,\cdot;U)$ always denotes the actual intrinsic metric in $U$, defined using lengths of $D$-continuous paths; a boundary restriction is not treated as a length metric on the boundary.

A coordinate complex is a finite connected union of compact coordinate rectangles of dimension $d-1$ with nonempty relative interiors. Rational complexes form a countable class containing cube boundaries and connected enlargements through any finite set of rational points. The proof uses only countably many deterministic complexes, similarities and scale fields. Their individual full-probability construction events are intersected when needed.

For the parameter argument we use mean-log normalization,

$$\rho_h^\zeta = e^{-\mathbb{E}\log D_h^\zeta(0,e_1)} D_h^\zeta, \qquad F_\zeta = \log(1+\rho_h^\zeta). \tag{1.5}$$

The logarithmic moments are proved in Section 5. These deterministic normalizations are kept fixed under field perturbations. Theorem A is invariant under deterministic multiplication, so its proof applies to either normalization. Section 10 returns to (1.1).

## Contents

**1 Introduction** **1**
**2 A scale decomposition of the field** **4**
**3 Separators at criticality** **5**
**4 Summability and Euclidean topology** **8**
**5 Uniform marginal estimates** **10**
**6 Comparison along geodesics** **15**
**7 Effective comparison inside a subcritical band** **19**
**8 Convergence on fixed surfaces** **23**
**9 A common spatial modulus** **25**
**10 Median normalization** **27**

## 2 A scale decomposition of the field

We use one decomposition throughout the paper. Its scale averages are Brownian motions; sphere averages in higher dimensions need not have this property. Choose a nonnegative radial $K \in C_c^\infty(\mathbb{R}^d)$ with $\int K^2 = 1$, and let $W$ be white noise on $\mathbb{R}^d \times (0,\infty)$ with Lebesgue control measure. Set

$$H_t(x) = \int_{e^{-t}}^1 \int_{\mathbb{R}^d} r^{-(d+1)/2} K((x-y)/r)\, W(dy,dr), \qquad t \ge 0. \tag{2.1}$$

Let $X$ denote the distribution obtained by integrating over $0 < r < 1$. If $k = K * \widetilde{K}$, then

$$\operatorname{Cov}(H_s(x), H_t(y)) = \int_0^{s\wedge t} k(e^u(x-y))\, du. \tag{2.2}$$

In particular, $H_\cdot(x)$ is standard Brownian motion. The fine field $X - H_t$ is independent of the noise at scales at least $e^{-t}$, has dependence range $O(e^{-t})$, and satisfies

$$(X - H_t)(z + e^{-t}\,\cdot) \overset{\text{law}}{=} X. \tag{2.3}$$

These are identities for distributions with their additive constants retained.

The large-scale increment

$$G(x) = \int_1^\infty \int_{\mathbb{R}^d} r^{-(d+1)/2} \{K((x-y)/r) - K(-y/r)\}\, W(dy,dr)$$

is locally smooth: every positive-order derivative has integrable variance at infinity, as does the displayed order-zero difference. Radiality and a change of variable in the scale integral show that $X + G$ has covariance $-\log|x-y|$ on zero-mass tests. After fixing the sphere pin, we therefore have

$$h = X + f, \qquad f \text{ locally smooth almost surely.} \tag{2.4}$$

The constant in $f$ can depend on $X$. The simultaneous continuous-perturbation theorem in [1, Proposition 13.7] permits this dependence.

### 2.1 Oscillations conditioned on the terminal height

Fix $R < \infty$, and define

$$R_n(z) = \sup_{0\le t\le n} \sup_{|w-z|_\infty \le Re^{-n}} |H_t(w) - H_t(z)|.$$

**Lemma 2.1.** *For each $M < \infty$ there are $c_1, C > 0$, uniform in $n \ge 1$ and $z$, such that*

$$\mathbb{P}[R_n(z) > u \mid H_n(z) = a] \le Ce^{-c_1 u^2}, \qquad |a| \le Mn, \quad u \ge C.$$

*Proof.* By (2.2) and $1 - k(v) = O(|v|^2)$, the covariance of $H_t(w) - H_t(z)$ with $H_n(z)$ is bounded by $C_R$, for $t \le n$ and $|w-z|_\infty \le Re^{-n}$. Since $\operatorname{Var} H_n(z) = n$, conditioning produces a mean of absolute value at most $C_R M$.

On the strip $n - j - 1 \le t \le n - j$, the variance of the difference process and its squared canonical metric, in coordinates $u = e^n(w - z)$, are bounded by $C_R e^{-2j}$ and

$$C_R e^{-2j}(|u-v|^2 + |t-s|),$$

respectively. Gaussian entropy on the fixed parameter box bounds the expected absolute supremum there by $C_R e^{-j}$. Summing over $j \geq 0$ bounds the expected supremum uniformly. Conditioning decreases the canonical metric of the centered residual process; Gaussian comparison gives the same bound for that process. Its maximum variance is also uniformly bounded. Gaussian concentration, together with the bound on its conditional mean, proves the assertion. □

Put $q = \sqrt{2d}$, fix $\kappa = 10^{-4}$, and, for a bounded box $J$, write

$$\mathcal{G}_n(J) = \kappa e^{-n}\mathbb{Z}^d \cap J, \qquad V_n(x) = qn - H_n(x).$$

**Lemma 2.2.** *For every $a_0 < 1/(2q)$, almost surely, for all sufficiently large $n$,*

$$V_n(z) \geq a_0 \log n \quad (z \in \mathcal{G}_n(J)). \tag{2.5}$$

*For every $M > q$, one also has $|H_n(z)| \leq Mn$ at all these points, eventually.*

*Proof.* The variance is $n$, and (2.2) gives

$$\mathbb{E}[(H_n(x) - H_n(y))^2] \leq 2 \log_+(e^n|x-y|) + C.$$

After indexing the grid by a lattice box of side $N \asymp e^n$, these are precisely the bounds in Assumption (A.0) of [4]. Proposition 1.1 of that paper bounds the upper tail relative to $qn - \frac{3}{2q}\log n + O(1)$ and gives

$$\mathbb{P}\left[\max_{z \in \mathcal{G}_n(J)} H_n(z) > qn - a_0 \log n\right] \leq C(\log n) n^{-(3/2 - qa_0)}.$$

This is summable. The last assertion follows from a Gaussian union bound over $O(e^{dn})$ points, since $M^2/2 > d$. Apply Borel–Cantelli to both bounds. □

## 3 Separators at criticality

In this section the metric has parameter $c = \zeta_c$, so that its similarity exponent is $cq$. We first obtain a quantitative separator estimate using only intrinsic continuity on deterministic cube boundaries.

**Lemma 3.1.** *If $C$ is a deterministic closed coordinate cube and $U$ is an open neighborhood of $\partial C$, the actual intrinsic metric on $U$ is finite and continuous on $(\partial C)^2$ and has finite diameter there. The statement remains true after a continuous perturbation of the field.*

*Proof.* Choose $\partial C \subset V \Subset U$. Lower semicontinuity and separation give

$$b = \inf\{D_h^c(x,y) : x \in \partial C,\ y \in \partial V\} > 0.$$

If $b = \infty$, replace it by any positive number. Uniform continuity on $(\partial C)^2$ supplies $\delta > 0$ such that $D_h^c(x,y) < b/2$ for $|x-y| < \delta$. A sufficiently short near-minimizing path cannot leave $V$, because finite metric paths are Euclidean continuous and an exit costs at least $b$. Thus the intrinsic and global distances agree for these pairs. Compactness and connectedness of $\partial C$ give chains with a uniformly bounded number of $\delta$-small steps. This proves a finite intrinsic diameter; the triangle inequality gives continuity. Weyl scaling on $V$ bounds the perturbed path lengths by $e^{c \sup_V f}$ times the original lengths, proving the final assertion. □

Use $B_r(z) = z + (-r, r)^d$ and fix $R = 200$. Let $\mathcal{S}$ be the countable collection of finite unions of rational cube boundaries contained in $B_2 \setminus \overline{B}_1$ and separating $\overline{B}_1$ from $\mathbb{R}^d \setminus B_2$. For the field $X$, put

$$T = \inf_{S \in \mathcal{S}} \operatorname{diam}_{D^c_X(\cdot,\cdot;B_R)} S, \quad J_0 = D^c_X(\partial B_{1/100}, \partial B_{100}; B_R), \quad Y = 1 + T + J_0. \tag{3.1}$$

All variables are measurable. Lemma 3.1 reduces boundary diameters and, when finite, cross-distances to countable dense sets. If a cross-distance is finite, finite boundary diameters make all cross-distances finite and continuous; otherwise their infimum is infinite.

**Proposition 3.2.** *There are $c_1, C, u_0 > 0$ such that*

$$\mathbb{P}[Y > e^u] \le C \exp\left(-c_1 \frac{u^2}{\log u}\right), \qquad u \ge u_0. \tag{3.2}$$

*Proof.* We give the lattice construction first. In $\mathbb{Z}^d$, use nearest-neighbor adjacency and $*$-adjacency, the latter meaning $\ell^\infty$ distance one. Label vertices good or bad, with all but finitely many good, and suppose each bad $*$-component has diameter at most $L$. For a finite $*$-connected $A$, add every bad component at distance at most one from $A$ to form $U(A)$. This set is $*$-connected and lies within distance $L+1$ of $A$. Its external $*$-boundary $\Gamma(A)$, visible from infinity along nearest-neighbor paths avoiding $U(A)$, consists of good vertices and is nearest-neighbor connected by [5, Theorem 4].

Associate to $v$ the tile $C_v = N^{-1}(v + [-1/2, 1/2]^d)$. Suppose each good tile has intrinsic boundary diameter at most $b$ in its doubled buffer $N^{-1}(v + (-2, 2)^d)$. The boundaries of any finite connected collection $\Gamma$ of good tiles have intrinsic diameter at most $2b\#\Gamma$: follow a simple nearest-neighbor route and connect shared-face contact points inside the successive buffers.

Assume $L \le 10^{-6}N$. Apply the construction to $A = \{|v|_\infty \le \lfloor 3N/2 \rfloor\}$. The union of tile boundaries indexed by $\Gamma(A)$ lies between $\partial B_1$ and $\partial B_2$ and separates them. Indeed, fill the finite components of $\mathbb{Z}^d \setminus U(A)$. The union of tiles in the resulting filled set contains $\overline{B}_1$, lies in $B_2$, and has Euclidean boundary covered by the tile boundaries indexed by $\Gamma(A)$. To check the last inclusion, an exterior tile touching the filled set is $*$-adjacent to it. If the adjacent tile is in a filled hole, a nearest-neighbor route through their common unit lattice cube must encounter $U(A)$ before reaching the exterior. The exterior vertex is therefore $*$-adjacent to $U(A)$ and belongs to $\Gamma(A)$.

For a crossing apply the construction to $A = \{0, e_1, \dots, 101Ne_1\}$. The external boundary contains a vertex within $L+2$ of zero and one beyond $101Ne_1$, by taking the first exterior vertices on the negative and positive $e_1$ rays. The first tile lies in $B_{1/100}$ for large $N$ and the second lies outside $B_{100}$. A route through the connected external boundary, truncated at these two surfaces, supplies the crossing. All buffers lie in $B_{103}$, and each boundary collection has at most $C_d N^d$ vertices. Both costs are therefore at most $C_d N^d b$.

Now take $N = \lceil u^2 \rceil$, $\ell = \log N$, and split $X = H_\ell + \widehat{X}_\ell$. A tile centered in $B_{150}$ is good when

$$\operatorname{diam}_{D^c_{\widehat{X}_\ell}(\cdot,\cdot;N^{-1}(v+(-2,2)^d))} \partial C_v \le A N^{-cq}.$$

Declare all other vertices good. Locality, (2.3), and exact similarities give a fixed finite dependence range in lattice units. Lemma 3.1 allows $A$ to make the bad probability $p$ arbitrarily small. A self-avoiding bad $*$-path of length $m$ contains $m/C_0$ sites with disjoint determining noise regions and hence has probability at most $p^{m/C_0}$. Counting at most $(3^d)^m$ paths per initial vertex gives

$$\mathbb{P}[\text{some bad component has diameter } > 10^{-6}N] \le C N^d e^{-c_2 N}.$$

On its complement the lattice construction and Weyl scaling imply

$$Y \le 1 + CAN^{d-cq} \exp\left( c \sup_{B_R} H_\ell \right). \tag{3.3}$$

The expected supremum of $H_\ell$ on $B_R$ is at most $C(1+\ell)$: cover by $O(e^{d\ell})$ cutoff-size boxes, use entropy inside each box, and a Gaussian union bound at their centers. Its maximum variance is $\ell$. Gaussian concentration gives

$$\mathbb{P}[\sup_{B_R} H_\ell > u/(2c)] \le C \exp(-c_3 u^2/\ell)$$

for large $u$. Since $\ell = O(\log u)$, the polynomial factor in (3.3) is at most $e^{u/4}$ eventually. These two failure estimates prove (3.2). □

For the rescaled fine field $X_{n,z}(u) = (X - H_n)(z + e^{-n}u)$, let $Y_{n,z}$ be (3.1), and set

$$Z_n(z) = (1+c)R_n(z) + \log(4Y_{n,z}).$$

The fine field is independent of the full coarse noise. Lemma 2.1 and Proposition 3.2 imply, uniformly for $|a| \le Mn$,

$$\mathbb{P}[Z_n(z) > u \mid H_n(z) = a] \le C \exp(-c_1 u^2/\log u). \tag{3.4}$$

Scaling a separator and a crossing from (3.1) gives a separator $S_n(z) \subset B_{2e^{-n}}(z) \setminus \overline{B}_{e^{-n}}(z)$ and a path $P_n(z)$ crossing from radius $e^{-n}/100$ to $100e^{-n}$, both in the buffer $B_{Re^{-n}}(z)$, whose intrinsic diameter and length are at most

$$e^{-cV_n(z)+Z_n(z)}. \tag{3.5}$$

The factors are $e^{-cqn}$ from scaling and at most $e^{cH_n(z)+cR_n(z)}$ from Weyl scaling. Approximate infima in (3.1); no measurable choice of paths is needed.

**Lemma 3.3.** *Fix $p \in (1/2, 1)$ and $0 < a_0 < 1/(2q)$. Almost surely, eventually at every mesh point in a fixed bounded box,*

$$V_n(z) \ge a_0 \log n, \qquad Z_n(z) \le V_n(z)^p.$$

*Proof.* Discard the summable exceptional events in Lemma 2.2. At a mesh point the density of $H_n(z) = qn - v$ is

$$(2\pi n)^{-1/2} \exp\{-dn + qv - v^2/(2n)\}.$$

Multiplication by the number of mesh points cancels $e^{-dn}$. Conditional on the height, (3.4) thus bounds the probability of any failure by

$$Cn^{-1/2} \sum_{v \ge a_0 \log n - 1} \exp\left( q(v+1) - c_1 \frac{v^{2p}}{\log v} \right).$$

After adjusting finitely many bins, this decays faster than any power of $n$, because $2p > 1$. Borel–Cantelli finishes the proof. □

For $x$ in a fixed box let $z_n(x)$ be a nearest point of $\kappa e^{-n}\mathbb{Z}^d$. Use a slightly larger box for the mesh. The definition of $R_n$ and Lemma 3.3 give

$$\sup_{t\le n} |H_t(x) - H_t(z_n(x))| \le V_n(z_n(x))^p. \tag{3.6}$$

Consequently, for every $a < a_0$,

$$V_n(x) \ge a \log n \tag{3.7}$$

uniformly in $x$, eventually. For every $\eta < c$, the costs in (3.5), centered at $z_n(x)$, are at most $e^{-\eta V_n(x)}$ eventually, since, with $v = V_n(z_n(x))$,

$$-cv + v^p + \eta V_n(x) \le -(c-\eta)v + (1+\eta)v^p \le 0.$$

# 4 Summability and Euclidean topology

The logarithmic barrier alone does not make the costs summable. We control the number of scales close to that barrier before joining the separators.

**Lemma 4.1.** *Fix $0 < a < b$. Let $N, n$ be integers with $N \ge N_0$ and $N \le n \le 2N$, where $N_0$ is a sufficiently large fixed integer. Let $(B_k)_{k=0}^n$ be the standard bridge from zero to zero over $[0, n]$, sampled at integer times, and let $a \log N \le v \le b \log N$. Put $U_k = kv/n - a \log N$ and $L_k = kv/n - b \log N$. For $s \ge 1$,*

$$\mathbb{P}[B_k \le U_k \ (N \le k \le n),\ \#\{k \in [N, n] : B_k \ge L_k\} \ge s] \le C \exp\{-c_1 s/(\log N)^2\}.$$

*Proof.* Let $L = \lceil (\log N)^2 \rceil$. At a stopping time $\tau \le n - 2L$ with $B_\tau \ge L_\tau$, the conditional mean of $B_{\tau+L}$ is

$$\frac{n-\tau-L}{n-\tau} B_\tau \ge -b \log N,$$

and its variance is $L(n-\tau-L)/(n-\tau) \ge L/2$. As $U_{\tau+L} \le (b-a)\log N$, the conditional probability of exceeding the upper barrier is at least a fixed $p_0 > 0$. Choose the first eligible visit, and then the first eligible visit at least $L$ steps after the previous one. There are at least $s/L - 3$ trials if there are $s$ visits: the final $2L$ times and skipped blocks account for the loss. Each barrier check occurs before the next trial. Iterated conditional probabilities bound the event by $(1-p_0)^{\max\{\lfloor s/L \rfloor - 3, 0\}}$, which proves the claim. □

**Proposition 4.2.** *For each fixed $B < \infty$, almost surely, for all sufficiently large dyadic $N$,*

$$\sup_{x \in J} \#\{k \in [N, 2N] \cap \mathbb{Z} : V_k(x) \le B \log N\} \le (\log N)^4.$$

*Proof.* If a point has any counted times, take the last such time $n$ and set $z = z_n(x)$. By (3.6), $V_n(z) - V_n(z)^p \le B \log N$, hence $V_n(z) \le 2B \log N$ for large $N$. The same oscillation bound holds at every earlier time, with error $O((\log N)^p) = o(\log N)$. Thus, after increasing $b > a > 0$, the counted times have $V_k(z) \le b \log N$, while (3.7) gives $V_k(z) \ge a \log N$ throughout $[N, n]$ and at $n$.

For fixed $(n, z)$, condition on $V_n(z) = v \in [a \log N, b \log N]$. The variables $H_k(z) - k(qn - v)/n$ form a Brownian bridge independent of $v$. The two deficit inequalities become the barrier and visit inequalities of Lemma 4.1. The endpoint probability is at most $C(\log N) n^{-1/2} e^{-dn} N^{qb}$. There

are $O(e^{dn})$ points at level $n$ and at most $N+1$ choices of $n$. The probability of a witness with $(\log N)^4$ visits is therefore at most

$$CN^{qb+1/2}(\log N)e^{-c_1(\log N)^2},$$

summable over dyadic $N$. This proves the uniform assertion by Borel–Cantelli. In particular, we have not conditioned a bridge on the global barrier or taken an uncountable union of exceptional events. □

**Corollary 4.3.** *If $\eta > 0$ and $0 < \theta < \eta/(2q)$, almost surely there is a finite $C$ such that*

$$\sup_{x\in J} \sum_{n\geq m} e^{-\eta V_n(x)} \leq Cm^{-\theta} \tag{4.1}$$

*for all sufficiently large $m$.*

*Proof.* Choose $a < 1/(2q)$ with $\eta a > \theta$, and $B$ with $\eta B > 1+\theta$. On $[N, 2N]$, the preceding proposition and (3.7) bound the sum by

$$(\log N)^4 N^{-\eta a} + (N+1)N^{-\eta B} \leq CN^{-\theta}.$$

Sum over dyadic blocks, beginning with the block containing $m$. □

*Proof of Theorem A.* Choose $\eta < c$ with $\theta < \eta/(2q)$. Write $r_n = e^{-n}$ and abbreviate the separator and crossing centered at $z_n(x)$ by $S_n(x), P_n(x)$. Both costs are at most $a_n(x) = e^{-\eta V_n(x)}$ eventually.

For $N \geq m$, join the inner endpoint $x_N$ of $P_N(x)$ to the outer endpoint of $P_m(x)$ through these finite chains. The center displacement is at most $\kappa(r_n + r_{n+1})$. With our choice of $\kappa$, the inner endpoint of $P_n(x)$ lies inside the enclosed cubes for both $S_n(x)$ and $S_{n+1}(x)$, and its outer endpoint lies outside both outer cubes. Euclidean continuity therefore forces the crossing to meet both separators. Follow $P_N(x)$ to $S_N(x)$, join to a contact of $P_{N-1}(x)$ with that separator, and continue. Buffered intrinsic connections with total excess at most $\sum_{n=m}^{N} a_n(x)$ give a finite path of length at most $C\sum_{n=m}^{N} a_n(x)$, contained in $B_{(R+1)r_m}(x)$, with $|x_N - x|_\infty \leq (\kappa + 1/100)r_N$.

If $|x-y|_\infty \leq r_m/4$, the two chains starting at $x_N, y_N$ meet the common separator $S_m(x)$ for large $N$: their initial points are inside its inner cube and their outer endpoints, at radius $100r_m$ about their own centers, are outside its outer cube. Joining their contacts gives

$$D_X^c(x_N, y_N) \leq C\left(\sum_{n=m}^{N} a_n(x) + \sum_{n=m}^{N} a_n(y)\right) \leq Cm^{-\theta}.$$

All paths stay in a fixed compact enlargement. By (2.4) and Weyl scaling, their $D_h^c$ lengths obey the same bound with a random constant. Lower semicontinuity as $N \to \infty$ gives the bound at $x, y$ themselves. This argument applies on one event to every pair, including field-dependent points; finiteness at the limiting endpoints was not presumed.

Subdivide a Euclidean segment into finitely many sufficiently small pieces to deduce finiteness of every distance and bounded metric diameter on compact boxes. Choosing $m$ within a fixed constant of $\log(1/|x-y|)$ proves the stated modulus. For the reverse topology, lower semicontinuity and separation give

$$b(x,\varepsilon) = \min_{z\in\partial B_\varepsilon(x)} D_h^c(x,z) > 0.$$

A finite path from $x$ to outside $B_\varepsilon(x)$ crosses its boundary and costs at least $b(x,\varepsilon)$. The length property gives $B^{D_h^c}_{b(x,\varepsilon)}(x) \subset B_\varepsilon(x)$. Both identity maps are continuous. Countable box exhaustion and a countable cofinal choice of exponents complete the simultaneous assertion. □

## 5 Uniform marginal estimates

Choose $I = [c_-, c]$ sufficiently short that, with $s = d - 1$,

$$\mu := \inf_I \phi(\zeta) > c\sqrt{2s}, \qquad c_- > 0. \tag{5.1}$$

All uniformities in this section concern probabilities or expectations at deterministic parameters. A simultaneous bound for the whole random family will be obtained in Section 9.

We first record fixed-parameter topology without using a continuous parameter version. For each fixed $0 < \zeta < c$, the proof of Theorem A applies with Weyl parameter $\zeta$ and similarity exponent $\phi(\zeta)$. The separator estimate uses the same positive surface gap, and its physical cost is bounded by

$$e^{-\phi(\zeta)n+\zeta H_n+Z_n} = e^{-\zeta V_n+Z_n} e^{-\zeta(Q_d(\zeta)-q)n} \le e^{-\zeta V_n+Z_n}.$$

The field occupation estimate and finite-chain argument are unchanged. Thus each such metric is finite and induces the Euclidean topology; its constants may depend on $\zeta$. This is the fixed-parameter fact needed for terminal access below.

**Proposition 5.1.** *In the normalization (1.5), the following estimates hold uniformly for deterministic $\zeta \in I$. For each coordinate complex $S \Subset U$, there are $\alpha_s, p > 0$ and $H_{\zeta,S,U}$ such that*

$$\rho_h^\zeta(x, y; U) \le H_{\zeta,S,U}|x - y|^{\alpha_s}, \quad x, y \in S, \qquad \sup_{\zeta\in I} \mathbb{E}H^p_{\zeta,S,U} < \infty. \tag{5.2}$$

*Each fixed annular crossing $Z_\zeta$ satisfies*

$$\sup_{\zeta\in I} \mathbb{P}[Z_\zeta < e^{-u}] \le C \exp\left(-c_1 \frac{u^2}{\log(2 + u)}\right). \tag{5.3}$$

*Unit point distances have uniformly bounded logarithmic moments of every finite order. Geodesics with endpoints in a fixed complex have a uniform positive-power confinement tail. For every fixed $\beta > \sup_{\zeta\in I}\{\phi(\zeta) + q\zeta\}$ there are $p, C, A > 0$ and random $K_{\zeta,R} \ge 1$ such that, for $R \ge 1$,*

$$\rho_h^\zeta(x, y) \ge K_{\zeta,R}^{-1}(|x - y| \wedge 1)^\beta \quad (x, y \in B_R), \qquad \sup_{\zeta\in I} \mathbb{E}K^p_{\zeta,R} \le C(1 + R)^A. \tag{5.4}$$

*Surface moment bounds have the same polynomial radius dependence for translates of fixed templates. More generally, this holds for uniformly bounded plaque counts and diameters, positive lower bounds on nonzero side lengths and buffer clearances, and bounded connecting-chain lengths with positive clearance, as in [1, Proposition 11.3].*

*Proof. Quantile normalization.* Start with any fixed normalization and set

$$a_\zeta = \mathrm{Quant}_{p_*} D^\zeta_X(0, e_1; B_2), \qquad M^\zeta_g = a_\zeta^{-1} D^\zeta_g,$$

where $p_* < 1$ is a fixed sufficiently high lower quantile. Separation and a buffered surface connection make $a_\zeta$ positive and finite. No bound on $a_\zeta$ is presumed. We have

$$\mathbb{P}[M^\zeta_X(0, e_1; B_2) \le 1] \ge p_*, \qquad \mathbb{P}[M^\zeta_X(0, e_1; B_2) \ge 1] \ge 1 - p_*. \tag{5.5}$$

The normalization is kept fixed for every field input. For $r = 2^{-\ell}$, exact scaling gives

$$M^\zeta_{X-H_{\ell\log 2}}(z + ru, z + rv; z + rU) \overset{\text{law}}{=} r^{\phi(\zeta)} M^\zeta_X(u, v; U). \tag{5.6}$$

*Surface chaining.* At level $\ell$, declare a site good if all its signed coordinate fine-field edges in fixed buffers cost at most $2^{-\ell\phi(\zeta)}$. Translations, orthogonal covariance [1, Corollary 20.10], (5.5), and a union bound give bad probability at most $2d(1-p_*)$, with fixed finite dependence range. The good-component chaining in [1, Lemmas 5.1–5.2] therefore has an eventual scale $N_\zeta$ satisfying

$$\sup_{\zeta\in I}\mathbb{P}[N_\zeta > n] \le Ce^{-c_1 n^2}. \tag{5.7}$$

Indeed, path counting rules out bad components of diameter larger than $\ell^2$ at all levels $\ell \ge N_\zeta$. Filling their finite holes gives compatible good vertices at consecutive levels and routes of at most $C(1+\ell)^{2d}$ edges in tubes of width $C(1+\ell)^2 2^{-\ell}$. A deterministic lower bound on $N_\zeta$ puts the tubes inside $U$.

Writing $T_\ell(S)$ for these tubes, the cost at mesh $2^{-m}$ is bounded by

$$CU_m(S), \qquad U_m(S) = \sum_{\ell\ge m}(1+\ell)^{2d}2^{-\mu\ell}\exp\{c\sup_{T_\ell(S)}|H_{\ell\log 2}|\}. \tag{5.8}$$

This common envelope depends only on the coarse fields. Terminal good vertices approach the requested endpoint Euclideanly. For each fixed parameter, Euclidean topology makes their metric distance tend to zero; at $c$ this is Theorem A. A positive exit barrier from a compact buffer confines the terminal connections to $U$. No uniform terminal modulus is used.

The tube has at most $C(1+\ell)^C 2^{s\ell}$ cutoff-size boxes. Gaussian entropy and a union bound give

$$\mathbb{E}e^{u\sup_{T_\ell(S)}|H_{\ell\log 2}|} \le C_u(1+\ell)^C 2^{s\ell+u^2\ell/2}.$$

Choose $\alpha_s, p_0 > 0$ with $p_0(\mu-\alpha_s) - s - p_0^2c^2/2 > 0$, possible by (5.1). Subadditivity if $p_0 \le 1$, or Minkowski otherwise, shows that $A_S = \sum_m 2^{\alpha_s m}U_m(S)$ has a finite $p_0$-moment. Chaining larger separations at level $N_\zeta$ gives $H_{\zeta,S,U} \le C2^{dN_\zeta}A_S$. Equation (5.7) and Hölder's inequality give a uniform positive moment. For a cube of diameter $O(2^{-m})$, retain the improved tube count $C(1+\ell)^C 2^{s(\ell-m)}$ to obtain

$$\mathbb{E}[U_m(\partial C)^{p_1}] \le C(1+m)^C 2^{-\nu m} \tag{5.9}$$

for fixed $p_1, \nu > 0$, uniformly over such deterministic cubes in a bounded region.

*Noncollapse.* Apply the zero-profile argument of [1, Section 6] to a sequence of continuum metrics. We verify its uniformity and limiting hypotheses. First, (5.5), polygonal chaining, and the uniform surface estimate give a uniform positive crossing witness as in [1, Lemma 6.1]. Finitely many point comparisons give a positive-probability separated point pair. Finite plaque covers with small patch diameters turn this into positive lower bounds for patch-pair infima. Positive association intersects these finitely many events. To justify association directly for the continuum assignments, approximate white noise by finite independent cell integrals at scales bounded away from zero. Increasing a coordinate adds a nonnegative smooth function because $K \ge 0$, so Weyl scaling makes intrinsic-distance lower events increasing. Conditional expectation over the residual noise preserves monotonicity. Product Gaussian association and martingale convergence prove association for the full field. Thus, for each fixed annulus,

$$\inf_{\zeta\in I}\mathbb{P}[M_X^\zeta(\text{across}) \ge a_0] \ge p_0' > 0. \tag{5.10}$$

If reciprocal tightness failed, choose $\zeta_j \to \zeta_*$ and a crossing limit $Z$ with positive mass at zero. Extract jointly the wired global and local source-distance profiles of [1, Section 6], on all rational polycubical domains, their versions with each fixed coarse band $H_a$ deleted, and the coarse fields.

These continuum wired profiles are path-length functionals of the actual local intrinsic metric, with the wired source assigned zero cost; their plaque restrictions are measurable by locality and countable approximation of endpoints. Plaque tightness follows from the uniform surface estimate. On the bounded region containing the nonwired costs,

$$e^{-c\|H_a\|_\infty} f^U_{i,j} \le f^{U,a}_{i,j} \le e^{c\|H_a\|_\infty} f^U_{i,j}.$$

Original and deleted zero sets consequently coincide in the limit. Deleted profiles are independent of each fixed coarse band; local deleted profiles in separated domains are independent by finite range and locality. These independence statements pass to the joint limit.

Retain also the common envelopes in (5.8)–(5.9) and their eventual good-scale events. Their joint law with the coarse fields is fixed, and (5.7) makes the limiting good events exhaustive. The deterministic first-exit and connectedness arguments of [1, Lemmas 6.2–6.3] apply to the zero clusters $E_1, E_2$. If they meet, their Borel-selected contact point is independent of the coarse fields. Conditioning on that point in (5.9) and applying Borel–Cantelli makes the surrounding separator costs tend to zero. Hence

$$\{Z > 0\} = \{E_1 \cap E_2 = \varnothing\} = \{E_1 \text{ avoids the opposite source}\} = \{E_2 \text{ avoids the opposite source}\}.$$

The independent-neighborhood calculation in [1, Proposition 6.4] gives $p \le p^2$ for $p = \mathbb{P}[Z > 0]$: sum factored probabilities over disjoint rational neighborhoods of the two clusters and let their mesh tend to zero; each marginal sum is at most $p$. Equation (5.10) and Portmanteau give $p > 0$, so $p = 1$, a contradiction. This proves uniform reciprocal tightness. Exact continuum scaling (5.6) replaces the cutoff-normalizer ratio argument in the source.

*Tails and normalization.* The finite-range amplification of [1, Lemma 8.1] now has common constants. At mesh $2^{-m}$ a good buffered fine shell has crossing at least $M^{-1}2^{-m\phi(\zeta)}$. Except with probability $C2^{dm}e^{-c_1 2^m}$, every macroscopic crossing visits a good shell and has cost at least

$$M^{-1}2^{-m\sup_I \phi} \exp\{-c \sup_V |H_{m\log 2}|\}.$$

Take $m = \lfloor 2\log_2 u \rfloor$ and use Gaussian concentration to prove (5.3) for $M^\zeta_X$. The correction in (2.4) has a Gaussian supremum tail on each buffer. Weyl scaling, Hölder's inequality and splitting according to $\|f\|_\infty \le u/(2c)$ transfer the surface moment and crossing tail to $M^\zeta_h$, without independence of $f$ and $X$.

A surface connection bounds $M^\zeta_h(0, e_1)$ above, and a fixed separating annulus bounds it below. Thus all logarithmic moments are uniformly bounded. The numbers $b_\zeta = e^{-\mathbb{E}\log M^\zeta_h(0,e_1)}$ are bounded above and below, and

$$b_\zeta M^\zeta_g = e^{-\mathbb{E}\log D^\zeta_h(0,e_1)} D^\zeta_g = \rho^\zeta_g.$$

The unknown $a_\zeta$ cancels. This proves the normalization and transfers the estimates.

Finally, exact scaling, Gaussian pin moments and Hölder's inequality give, for small fixed $t > 0$,

$$\sup_{\zeta\in I} \mathbb{E}[\rho^\zeta_h(\partial B_R, \partial B_{2R})^{-t}] \le CR^{-t\mu + Ct^2c^2} \le CR^{-\kappa_1}. \tag{5.11}$$

A geodesic with endpoints in $S$ has length bounded by a surface connection. Comparing this positive-moment bound to a mandatory large-radius crossing gives the confinement tail.

For completeness, choose $\epsilon > 0$ so that $\phi(\zeta) + (q+\epsilon)\zeta + \epsilon < \beta$ throughout $I$. At mesh $r = 2^{-m}$, a fixed finite family of grid-shell templates separates every pair whose Euclidean separation is

comparable to $r$. There are $O((1+R)^d r^{-d})$ relevant centers in $B_{R+1}$. Put $g_{z,r}(x) = h(z+rx) - h_r(z)$; its marginal law is that of the pinned field. For each shell template $\mathcal{A}$, exact similarity gives

$$\rho_h^\zeta(\text{across}\,(z+r\mathcal{A})) = r^{\phi(\zeta)} e^{\zeta h_r(z)} Z_{\zeta,z,r,\mathcal{A}}, \quad Z_{\zeta,z,r,\mathcal{A}} = \rho_{g_{z,r}}^\zeta(\text{across}\,\mathcal{A}).$$

The re-pinned crossing is at least $e^{-u} r^\epsilon$ outside probability $C_P e^{-Pu} r^{P\epsilon}$, for any fixed $P$, by (5.3). No independence from the pin is needed. The pin has variance at most $\log(1/r) + C\log(2+R) + C$. A Gaussian exponential bound, with slope $q+\epsilon$ and an additional $C\log(2+R)$ allowance, gives

$$h_r(z) \geq -(q+\epsilon)\log(1/r) - u - C\log(2+R)$$

at all these centers and scales, outside probability $C(1+R)^A e^{-p_1 u}$. Indeed, the spatial count is absorbed by $(q+\epsilon)^2/2 > d$; choose $P\epsilon > d$ for the fine crossings and sum both bounds over $m$. On the resulting event every such physical crossing is at least $(2+R)^{-C} e^{-(1+c)u} r^{\phi(\zeta)+(q+\epsilon)\zeta+\epsilon}$. First exit through a shell proves (5.4); the scale-one shells handle separations larger than one. The countable shell family defines a measurable constant satisfying

$$\mathbb{P}[K_{\zeta,R} > C(2+R)^B e^{(1+c)u}] \leq C(2+R)^A e^{-p_1 u}, \qquad u \geq 0.$$

For $0 < p < p_1/(1+c)$, integration gives $\mathbb{E}K_{\zeta,R}^p \leq C(2+R)^{A+Bp}$, uniformly in each deterministic parameter. The same grid count and Gaussian pin calculation give the asserted radius dependence for geometrically controlled surface templates. □

**Proposition 5.2.** *On the original field space,*

$$\rho_h^\zeta \longrightarrow \rho_h^c \quad \textit{in probability in } C_{\mathrm{loc}}((\mathbb{R}^d)^2) \quad \textit{as } \zeta \uparrow c. \tag{5.12}$$

*Proof.* Proposition 5.1 makes the boundary quantile and hence the separator tail (3.2) uniform in deterministic $\zeta$. The physical unit at depth $n$ is

$$e^{-\phi(\zeta)n + \zeta H_n(x)} = e^{-\zeta V_n(x)} e^{-\zeta(Q_d(\zeta)-q)n} \leq e^{-\zeta V_n(x)}. \tag{5.13}$$

The height-bin estimate of Lemma 3.3 has summable failures uniformly in each deterministic parameter. The maximum and occupation estimates are field-only statements. Choose a deterministic starting scale, a high quantile of their summability constant, and a high quantile of the correction in (2.4). Finite-chain gluing gives

$$\limsup_{r\downarrow 0} \sup_{\zeta\in I} \mathbb{P}\left[\sup_{x,y\in K,\ |x-y|\leq r} \rho_h^\zeta(x,y) > \varepsilon\right] = 0.$$

Finite subdivisions of Euclidean segments give tight compact diameters. Arzelà–Ascoli proves profile tightness.

Let $\zeta_j \uparrow c$ and extract a joint limit $(h, E)$ of $(h, \rho_h^{\zeta_j})$. A finite family of annuli separates every pair in a compact set at a prescribed positive Euclidean separation. Uniform reciprocal crossing tightness therefore makes the limiting continuous pseudometric separating. Equation (5.11), at rapidly growing deterministic radii, gives diverging barriers in the limit. The compact path argument of [1, Lemma 12.5] then confines bounded-cost paths and gives subsequential Euclidean path limits. Their arclength identities give limiting geodesics. Thus $E$ is a finite proper geodesic metric.

We verify the coupled assignment before using uniqueness. Retain buffered intrinsic arrays on rational complexes, field averages, and smooth Gaussian regression corrections in the joint extraction. The recovery arguments of [1, Lemmas 12.6–12.10] identify the actual intrinsic metric of $E$. For upper recovery, subdivide a limiting path into finitely many short pieces and use positive compact exit barriers to confine short prelimit links. For lower recovery, take compact limits of bounded-length paths in smaller buffers, then exhaust.

If retained corrections satisfy $f_j \to f$ locally uniformly, then

$$e^{\zeta_j f_j} \longrightarrow e^{cf} \tag{5.14}$$

uniformly on compacts, with positive upper and lower bounds there. The weighted recovery proof of [1, Lemma 12.10] applies to these varying weights. For conditional locality, use the Gaussian regression decomposition in [1, Proposition 13.3]. First re-pin inside the conditioning region $V$: choose a smooth probability density $\eta_V$ supported there and write $h^V = h - T$, where $T = h(\eta_V)$. Retain $T$ and the re-pinned metrics $e^{-\zeta_j T}\rho_h^{\zeta_j}$ in the extraction. After subtracting the smooth exterior correction, the exterior field is independent of the interior field. Prelimit local metric coordinates are functions of their respective fields, so the entire blocks are independent before passage to the limit. Retaining the regression fields and corrections preserves this independence, and (5.14) identifies the corrected metric. Return to the sphere pin by [1, Remark 4.7], retaining the scalar $T$; its Weyl factor converges to $e^{cT}$. This proves conditional locality without a general preservation claim for conditional independence under weak convergence.

At each fixed center and radius the units converge to $r^{cq}e^{ch_r(z)}$. The uniform surface and crossing quantiles pass to the retained arrays, giving regular weak estimates in these units. Apply the resampling argument of [1, Propositions 13.4–13.6], as in [1, Lemma 17.1, Steps 3–4]. Conditional locality gives product conditional kernels in disjoint cells, and regularity gives deterministic comparison of two conditional metric samples. The cell-resampling effect on a rational-pair distance is bounded by a constant times the geodesic arclength in that cell. This finite measure is atomless, so its largest cell mass tends to zero. The conditional Efron–Stein bound makes the variance vanish. Rational distances determine the continuous profile, hence $E$ is a Borel function of $h$. Every conditional sample compared in this step has the limiting parameter $c$.

The graph law gives convergence in probability for this subsequence on the original field space. Take a further subsequence with almost sure convergence of the global profiles and their retained surface restrictions. No convergence of every fixed-open-domain intrinsic array is needed. Use the continuum-sequence reconstruction of [1, Lemma 17.1, Step 4], which extends [1, Proposition 13.7]. On a fixed countable family of buffers and rational endpoints, its coordinates are

$$\mathcal{E}_U(g;p,q) = \inf_{V \Subset U} \limsup_j \rho_g^{\zeta_j}(p,q;V).$$

Here the prelimit coordinates use Borel compact-constraint versions; rational envelopes and the invalid-input convention in that reconstruction define a single Borel assignment on arbitrary distributional inputs. Deterministic pathwise recovery and (5.14) identify this assignment with the actual intrinsic metric of $e^{cf} \cdot E_h$, on one good base-field event for every continuous $f$, including dependent perturbations. For each fixed law of such a perturbation and each open $U$, the prelimit coordinates are measurable from $(h+f)|_U$; their countable formula has the required locality. No pointwise off-support locality assertion is needed. Retaining deterministic translates passes translation covariance; the limiting scale units are $r^{cq}$. Thus the limit is a regular weak assignment with parameters $(c,q)$.

Weak uniqueness [1, Theorem 19.1] gives $E_h = k\rho_h^c$ for a deterministic $k > 0$. Uniform logarithmic integrability and (1.5) give $\mathbb{E}\log E_h(0, e_1) = 0$, hence $k = 1$. Every joint subsequential limit is the same graph law. Joint tightness and the diagonal argument of [1, Proposition 20.8] give convergence in probability on the original field space. The initial sequence was arbitrary, proving (5.12). □

# 6 Comparison along geodesics

Fix the interval $I = [c_-, c]$ from Proposition 5.1, a coordinate complex $S$, and two deterministic parameters $\zeta, \eta \in I$. Write $A = \rho_h^\zeta$, $B = \rho_h^\eta$ and $\tau = |\zeta - \eta|$. Constants in this section are uniform in the two parameters. The positive surface exponent in Proposition 5.1 will replace the vanishing bulk exponent.

We use the annular Gaussian estimate of [2, Lemma 2.3], whose hypotheses concern only the field. For completeness, the zero-mass Gaussian spaces of an inner radius-$r$ window and an outer radius-$R$ annulus have operator correlation $O(r/R)$. Subtracting the constant from the logarithmic kernel and using Sobolev duality gives this estimate. At sufficiently separated geometric radii, the sum of off-diagonal operator norms in each row is at most one. The Gaussian product inequality then bounds the probability of simultaneous block events by the product of the square roots of their probabilities. Consequently, if a fixed-shape local annular event has sufficiently small failure probability, a block of $O(\log r^{-1})$ radii supplies a fixed positive density of good shells around every grid center in $B_W$, except with probability

$$C(1 + W)^C r^m. \tag{6.1}$$

Here $m$ can be prescribed before choosing the fixed reference quantiles. A union bound over a finest grid and fixed enlargements proves the assertion for every point, without a union over random paths.

## 6.1 Regularity on the source geodesic

**Lemma 6.1.** *There are $\chi, p > 0$ such that, for each $W \geq 1$, a random $H_W \geq 1$ satisfies $\mathbb{E}H_W^p \leq C(1 + W)^C$, and every $A$-geodesic $P : [0, L] \to B_W$ with endpoints in $S$, parametrized by $A$-arclength, satisfies*

$$|t - s| \leq H_W |P(t) - P(s)|^\chi. \tag{6.2}$$

*The same conclusion holds with $A, B$ interchanged.*

*Proof.* Choose fixed annular references so that each successful shell has source separator diameter at most $\Lambda$ times its source crossing. Suppose $J$ nested shells have disjoint enlarged annuli and avoid both endpoints of a geodesic. If $X_j$ denotes their crossing costs, ordered from outside inward, geodesicity gives

$$\sum_{\ell > j} X_\ell \leq \Lambda X_j.$$

Indeed, the full first-to-last visit span on the $j$th separator is no longer than its diameter, and includes the crossings of all smaller shells. Writing $T_j = \sum_{\ell \geq j} X_\ell$ gives $T_{j+1} \leq \Lambda T_j/(1 + \Lambda)$. The full innermost visit span is at most $\Lambda(\Lambda/(1 + \Lambda))^{J-1} L$. This is the ordinary-geodesic case of [1, Lemma 15.5].

For a candidate ball of radius $r$, use shells with radii between $r^{1/2}$ and $r^{1/4}$. On the density event (6.1), if both endpoints are outside the largest shell, the full visit span to the candidate ball is at most $Cr^{\chi_0}L$, with a fixed $\chi_0 > 0$.

If an endpoint $x \in S$ is within $O(r^{1/4})$ of the candidate ball, choose a good shell near $x$ with radius between $r^{1/8}$ and $r^{1/16}$ whose inner cube contains that ball. The finite rectangle geometry of $S$ supplies $y' \in S$ outside its outer cube and within $O(r^{1/16})$ of $x$. A surface connection from $x$ to $y'$ costs at most $CH_S r^{\alpha_s/16}$, where $H_S$ has a uniform positive moment and $\alpha_s > 0$ is the surface exponent. This path crosses the shell. Its prefix accesses the separator, and its full length bounds the source crossing. Every separator point is thus at distance at most $C(1+\Lambda)H_S r^{\alpha_s/16}$ from $x$. If the other endpoint is outside the outer cube, geodesicity bounds the prefix through the last separator contact by this cost; no later visit to the candidate ball occurs. If the other endpoint is inside the outer cube, both endpoints belong to $S$ at Euclidean distance $O(r^{1/16})$, so the surface bound controls $L$ itself. This is the endpoint-access argument of [1, Lemma 15.7], with its powers retained.

Choose $\chi < \min(\chi_0, \alpha_s/16)$. The surface estimate also bounds $L$ for all endpoints in $S$. At each dyadic radius the simultaneous full-span bound fails with probability at most (6.1). The last failed dyadic level $K$ has an exponentially decaying tail, with a polynomial prefactor in $W$. Hence $2^{\chi K}$ has a positive moment. Multiplying this factor by $CH_S$ and applying Hölder's inequality gives a constant $H_W$ with the stated moment. A fixed enlargement of the candidate balls covers every pair of image points. This proves (6.2) simultaneously for all the geodesics. □

## 6.2 Weighted routing to exact endpoints

The next argument keeps track of the cost of reaching an interior point of a source geodesic in the target metric. Small diameters of nested boundaries alone would not give such a connection.

For a fixed depth $n$, with $\log(1+W) \le c_0 n$, Gaussian union bounds give

$$|h_{r_k}(z)| \le Ck$$

at all required centers and all depths $k \ge n$, outside probability $Ce^{-c_1 n}$. There are $O((1+W)^d e^{Ck})$ centers at depth $k$, and the variance is $O(k+\log(1+W))$; choose the cap constant large and sum over $k$.

The similarity exponent is Lipschitz through the endpoint. Indeed, for cutoff LFPP distances $L_n^\zeta$ in a fixed buffer,

$$|\log L_n^\zeta - \log L_n^\eta| \le |\zeta - \eta| \sup |H_n|.$$

The supremum on that buffer is at most $Cn$ with probability tending to one. Divide by $n$ along integer multiples of $\log 2$ and use the deterministic distance exponents, valid also at $c$ [6, Proposition 1.1 and Lemma 2.7], to obtain $|\phi(\zeta) - \phi(\eta)| \le C|\zeta - \eta|$ on $I$. Thus the exact physical scale units satisfy

$$\frac{r^{\phi(\eta)} e^{\eta h_r(z)}}{r^{\phi(\zeta)} e^{\zeta h_r(z)}} \le Cr^{-C\tau}. \tag{6.3}$$

This bound is allowed to grow at arbitrarily fine scales.

Choose source crossing lower and target boundary diameter upper quantiles in their own units. Apply (6.1) and sum over dyadic $v \le e^{-n}$. On an event with failure $Ce^{-c_1 n}$, every point of $B_W$ and every such $v$ has a supplied cube with the point in its central core and radius $r \in [v^b, v]$, for a fixed $b > 2$, such that

$$\operatorname{diam}_B(\partial C; \text{buffer}) \le \Lambda r^{-C\tau} A(\text{inner boundary}, \text{outer boundary}; \text{annulus}). \tag{6.4}$$

Fixed geometric constants are absorbed by increasing $b$.

**Lemma 6.2.** *On this event, suppose (6.2) holds with constant $H$. For every subsegment $P : [0, \ell]$ of an original source geodesic with endpoints in $S$, and $\alpha = bC\tau/\chi \le 1/2$,*

$$B(P(0), P(\ell)) \le C\{e^{bC\tau n}\ell + H^{\alpha}\ell^{1-\alpha}\}. \tag{6.5}$$

*Proof.* Set $m(t) = \min(t, \ell - t)$ and choose a supplied cube at $P(t)$ using

$$v(t) = \min\{e^{-n}, \kappa_0(m(t)/H)^{1/\chi}\}, \qquad 0 < t < \ell.$$

Apply the supply at a dyadic $v_t \in [v(t)/2, v(t)]$, so $(v(t)/2)^b \le r \le v(t)$. Here $\kappa_0$ is a sufficiently small fixed constant. By (6.2), the cube and its buffer avoid both true endpoints with clearance proportional to its radius. The full source visit span to a fixed enlargement of the cube is at most $CHr^{\chi} \le m(t)/4$. Thus $m(s)$ is comparable to $m(t)$ on every charged time interval. Dyadic rounding changes the weight only by the bounded factor $2^{bC\tau}$; the factor in (6.4) is bounded there by

$$C\{e^{bC\tau n} + H^{\alpha}m(s)^{-\alpha}\}.$$

Apply the finite boundary-routing construction of [1, Lemma 12.4] to $P|_{[\varepsilon,\ell-\varepsilon]}$. Radii on this truncation have a positive lower bound. Select successive core visits, charge their source crossings on disjoint time intervals, remove contained cubes, and route through the boundaries along a simple path in the remaining intersection graph. Nonnested intersecting cubes have intersecting boundaries, and each retained separator is charged at most once. Integrating the preceding weight over the charged intervals bounds the routed cost by

$$C\int_0^{\ell}\{e^{bC\tau n} + H^{\alpha}m(t)^{-\alpha}\}\,dt \le C\{e^{bC\tau n}\ell + H^{\alpha}\ell^{1-\alpha}\}.$$

The bound is independent of $\varepsilon$.

The routed endpoints converge to the true endpoints. Indeed, a maximal retained cube containing $P(\varepsilon)$ still avoids $P(0)$ with clearance proportional to its radius. Its radius is at most $C|P(\varepsilon) - P(0)|$, and its boundary contact therefore tends to $P(0)$. The same holds at the other endpoint. If a single cube contains both truncated endpoints, this observation handles the constant-path alternative; for a fixed nonzero segment it is eventually impossible. Lower semicontinuity of $B$ now proves (6.5). No target length was assigned to the source geodesic. □

If $\tau n \le c_0$ and $H \le e^{hn}$, the two exponential factors in (6.5) are bounded. For $\ell \ge e^{-b_0 n}$, the additional factor $\ell^{-\alpha}$ is also bounded; for smaller $\ell$ the right side is at most $Ce^{-b_0 n/2}$. Consequently there are fixed $C_b, c_2 > 0$ such that

$$B(P(s), P(t)) \le C_b(t - s) + e^{-c_2 n} \tag{6.6}$$

on every subsegment of every original surface-ended source geodesic. Confinement from Proposition 5.1 and the moment in Lemma 6.1 make this an exponentially high-probability statement: take $W = e^{wn}$ and $H = e^{hn}$, choosing $w$ sufficiently small after $h$. Repeat for the reverse orientation.

### 6.3 Amplification of a probabilistic seed

**Proposition 6.3.** *For each fixed rational coordinate complex $S$, there are $A_0, M, C > 1$ and $c_0, p_0, \delta_0 > 0$ such that the seed*

$$\mathbb{P}[\|\rho_h^\zeta - \rho_h^\eta\|_{K_{A_0}} > c_0\delta] \le p_0 \tag{6.7}$$

*implies*

$$\mathbb{P}[\|F_\zeta - F_\eta\|_S > \delta] \le C\delta^{-M} e^{-c_0\delta^M n} \tag{6.8}$$

*whenever $0 < \delta < \delta_0$, $n \ge C\delta^{-M}\log(C/\delta)$ and $\tau n \le c_0\delta$.*

*Proof.* Choose local annular references before $\delta$: source separator upper bounds, source crossing lower bounds, and exit barriers for both metrics. More precisely, fix a compact middle annulus $K$, an open guard $V$ and a larger annular inspection $U$ with $K \subset V \Subset U$, including the local averaging sphere in $U$. For $R = A, B$, use the local barrier $\beta_R = R(K, \partial V; U)$. Proposition 5.1 gives fixed positive lower quantiles for both barriers, with arbitrarily small failure. A source geodesic visiting the core has an outer-boundary span of source length at most $M_0$ in its own units, by the separator competitor. Trim between the last contact with the inner boundary and the first subsequent contact with the outer boundary, so the middle crossing stays in $K$. It has length between $m_0$ and $M_0$. Split it into a fixed $N$ equal-arclength pieces, with $M_0/N$ less than half both barrier quantiles. Each piece has length at least $m_0/N$. Any cheaper ambient path would remain in $V$, since its first exit already costs $\beta_A$. Thus its source inspection distance equals its ambient distance under every exterior extension. On the seed event its target endpoint distance is at most its source length plus $c_0\delta$, hence at most $1 + O(\delta)$ times that length. Choose $c_0\delta$ below the remaining target-barrier margin. Target near-minimizers then stay in $V$ and give the required inspection-intrinsic connection.

The local predicate quantifies over source paths in $K$ which are also minimizing for the actual intrinsic metric of $U$. This requirement excludes cheaper routes elsewhere in the inspection, and every piece of an actual global source geodesic satisfies it. The preceding implication defines a local annular predicate applying to every required thin block. For measurability, use Borel compact-constrained coordinates and a countable compact exhaustion for the actual inspection metric, as in [1, Lemmas 12.2–12.3 and Section 14]. Existence of a violating path is an analytic event in the local metric and path data, so the predicate is measurable in the completed local sigma field. This does not require a uniform bulk upper modulus. The seed only bounds the predicate's probability. Fixed pin changes are paid by a fixed-confidence Gaussian cap; $\tau \le c_0\delta$ makes their multiplicative error $O(\delta)$. Thus the predicate has failure at most the fixed threshold required in (6.1). In particular it remains annular, so separated-scale decorrelation applies.

We spell out the sewing calculation. Suppose the current comparison on all subsegments of one original source geodesic is $B \le C_j A + e_j$. In a band of radii $[e^{-4a_j n}, e^{-a_j n}]$, cover every point by a good core. An inclusion-minimal interval cover has multiplicity at most two. At most two intervals are clipped by the subsegment endpoints; their total source length is at most $CHe^{-\chi a_j n}$ by Lemma 6.1. Each remaining interval has a fixed fraction of its source length in a seeded thin block. Color the intervals with two colors and retain the color carrying more block length. Its disjoint thin blocks have total length at least

$$\vartheta\{\ell - CHe^{-\chi a_j n}\}$$

for a fixed $\vartheta > 0$. A finite Gaussian cap gives every retained block source length at least $e^{-B_0 a_j n}$: its own-unit crossing has a fixed positive lower bound, and both its scale depth and the logarithm of the confinement radius are $O(a_j n)$. The number of blocks is therefore at most $\ell e^{B_0 a_j n}$.

Use the seeded connectors on these blocks and the old exact-endpoint comparison on their gaps. Every gap endpoint remains on the same original source geodesic, so (6.6) and all subsequent improvements apply. With $b = 1 + O(\delta)$, including the physical-unit drift, the resulting recursion is

$$\begin{aligned} C_{j+1} &\le b + (1-\vartheta)(C_j - b) + Ce_j e^{B_0 a_j n}, \\ e_{j+1} &\le CC_j H e^{-\chi a_j n} + e_j. \end{aligned} \tag{6.9}$$

If $C_j < b$, retain the existing comparison. This is the interval-sewing proof of [2, Lemma 3.1], with the trace exponent replacing its bulk upper exponent.

Start from (6.6). Choose $a_1$ small compared with $c_2/B_0$, then $L > 4B_0/\chi$, and set $a_j = a_1 L^{-(j-1)}$. Use $N = \lceil C_\vartheta \log(1/\delta) \rceil$ bands, so $(1-\vartheta)^N (C_b + 1) \le \delta/100$. Their smallest relative depth satisfies $a_* = a_N \ge c\delta^{M_0}$. Choose confinement and trace thresholds $W = Ce^{wa_* n}$, $H = Ce^{ha_* n}$, first making $h$ small compared with $\chi, c_2$ and then $w$ small compared with the moment exponent. The total failure of these thresholds, the annular covers and Gaussian caps is at most $C\delta^{-M} e^{-c_3 \delta^M n}$.

The first error quotient in (6.9) is $O(e^{-(c_2 - B_0 a_1)n})$. Later quotients are bounded by

$$CHe^{-(\chi a_j - B_0 a_{j+1})n} \le CHe^{-3\chi a_j n/4}.$$

The preceding errors form a geometric sum because the physical scales increase. Increasing $M, C$ makes the sum of quotient errors and the last additive error at most $\delta/100$ under the stated lower bound on $n$. Finite-depth unit ratios are $e^{O(\tau n)}$, so the drift is paid by $\tau n \le c_0 \delta$. Allocating the fixed constants gives $B \le (1 + \delta/2)A + \delta/2$ on $S^2$. Apply the argument in reverse. Since $1 + B \le (1 + \delta/2)(1 + A)$ and conversely, the logarithmic profiles differ by at most $\delta$. This proves (6.8). □

## 7 Effective comparison inside a subcritical band

Write $T_0(u) = u$ and $T_{j+1}(u) = e^{T_j(u)}$. We need a weak quantitative bound whose tower height remains fixed as the band approaches criticality.

**Proposition 7.1.** *For each fixed compact cube $S = K_R$, and for each fixed rational coordinate complex $S$, there are fixed integers $k, C_1 \ge 1$ and $a_0 > 0$ such that, with*

$$I_a = [c - 2a, c - a], \qquad \mathcal{R}(a, \delta) = T_k\big(C_1 (a\delta)^{-C_1}\big),$$

*one has*

$$\mathbb{P}[\|F_\zeta - F_\eta\|_S > \delta] \le \mathcal{R}(a, \delta) \exp\{-n/\mathcal{R}(a, \delta)\} \tag{7.1}$$

*for $\zeta, \eta \in I_a$, $0 < a < a_0$, $0 < \delta < \delta_0$, whenever*

$$n \ge \mathcal{R}(a, \delta), \qquad |\zeta - \eta| n \le \mathcal{R}(a, \delta)^{-1}.$$

*Proof.* We track the constants of [2, Sections 3–4]. Two preliminary estimates make the dependence on $a$ explicit. The lower parameter cutoff denoted by $a$ in that paper is fixed here at $c_- > 0$. Our $a$ is the distance of the band from criticality; its deterioration is tracked through the bulk gap and the subsequent choices below. The uniform Lipschitz bound for $\phi$ proved in Section 6 is part of the fixed data, so the physical-unit drift constants do not deteriorate with $a$.

*The bulk gap.* For the stationary cutoff field $H_n$, let $L_n^\zeta$ be a buffered unit point LFPP distance. Its deterministic exponent is $n^{-1}\log L_n^\zeta \to \phi(\zeta)-1$ in probability [6, Proposition 1.1 and Lemma 2.7]; it suffices to take $n$ through integer multiples of $\log 2$. For $\zeta<\eta$, take an independent $H_n'$ and couple

$$\eta\widetilde{H}_n = \zeta H_n + \sqrt{\eta^2-\zeta^2}\, H_n'.$$

Choose an $H_n$-measurable polygonal path with source length at most $2L_n^\zeta$. Conditional expectation of its target length multiplies its source length by $e^{(\eta^2-\zeta^2)n/2}$. Markov's inequality and the two deterministic exponents give, on letting the extra exponential tolerance decrease to zero,

$$\phi(\eta)-\phi(\zeta)\le(\eta^2-\zeta^2)/2.$$

The point-exponent bound $Q_d(\zeta)\le\zeta^{-1}+\sqrt{2}$ in [6, Proposition 1.1, (1.9)] gives $c\le(q-\sqrt{2})^{-1}<q$. Taking $\eta=c$ yields

$$\phi(\zeta)-q\zeta\ge(c-\zeta)\left(q-\frac{c+\zeta}{2}\right)\ge(q-c)(c-\zeta). \tag{7.2}$$

Thus the bulk upper exponent on $I_a$ can be chosen at least a fixed multiple of $a$.

*Effective regularity quantiles.* The continuum fine-edge quantiles are already uniform by Proposition 5.1; no rate of cutoff convergence is needed. Take $\epsilon$ a sufficiently small fixed multiple of $a$, so $\phi(\zeta)-(q+\epsilon)\zeta\ge c_2a$ on $I_a$. Cover $B_R$ at cutoff $e^{-n}$ by $O((2+R)^de^{dn})$ cells. The center height is $N(0,n)$, and the within-cell oscillation has bounded mean and a uniform Gaussian tail. Split the threshold $(q+\epsilon)n+u$ between center level $(q+\epsilon/2)n+u/2$ and oscillation $\epsilon n/2+u/2$. Summing the center tails over $n$ gives $C(2+R)^d\epsilon^{-1}e^{-qu/2}$. For the oscillations, use $(\epsilon n+u)^2\ge\epsilon^2n^2+u^2$ to obtain $C(2+R)^de^{C/\epsilon^2}e^{-c_3u^2}$. Consequently

$$Z_R=\left[\sup_{n\ge1,x\in B_R}\{H_n(x)-(q+\epsilon)n\}\right]_+$$

has a fixed positive exponential moment bounded by $C(2+R)^Ae^{C/a^2}$.

The good-component construction has a last failed scale $M_R$ with tail $C(2+R)^de^{-c_3m^2}$, independently of $a$. A good edge at level $n$ costs at most $Ce^{cZ_R}e^{-c_2an}$, with a fixed polynomial number of edges in each chain. Since

$$\sum_{n\ge j}(n+1)^Ke^{-c_2an}\le Ca^{-(K+1)}e^{-c_2aj/2},$$

the buffered bulk Hölder constant is bounded by $Ca^{-C}\exp(CM_R+cZ_R)$. The smooth correction in (2.4) satisfies $\mathbb{E}e^{\lambda\|f\|_{B_R}}\le C_\lambda(2+R)^{C_\lambda}$ for every fixed $\lambda>0$: the integral defining $G$ gives point variance $O(\log(2+R))$ and bounded unit-cell oscillations, and a union over $O((2+R)^d)$ cells gives the bound. The moment calculation and Hölder's inequality therefore give

$$\sigma\ge c_4a,\qquad \mathbb{E}[(H_{R,\zeta}^+)^p]\le C(2+R)^A\exp(Ca^{-C}), \tag{7.3}$$

for a fixed $p>0$. The inverse spatial exponent $\beta$ and inverse crossing moments remain fixed, by Proposition 5.1. To write its inverse estimate without the truncation $|x-y|\wedge1$ on $B_R$, multiply $K_{\zeta,R}$ by $(2\sqrt{d}R)^\beta$; its moment still grows polynomially in $R$.

The comparison also needs confinement for all endpoint pairs in a ball. Set

$$J_{R,\zeta}=\sup_{x,y\in B_R}\rho_h^\zeta(x,y;B_{2R}).$$

Finite chaining of (7.3) gives $\mathbb{E}J_{R,\zeta}^{p} \le C(1+R)^{A}e^{Ca^{-C}}$. Let $Z_{W,\zeta} = \rho_h^{\zeta}(\partial B_W, \partial B_{2W})$, where $W \ge 4R$. Equation (5.11) gives fixed $t, \kappa_1 > 0$ with $\mathbb{E}Z_{W,\zeta}^{-t} \le CW^{-\kappa_1}$. For any factor $K \ge 1$, put $\lambda = \kappa_1/(2t)$ and split at $W^{\lambda}$ to obtain

$$\mathbb{P}[KJ_{R,\zeta} \ge Z_{W,\zeta}] \le CK^{p}(1+R)^{A}e^{Ca^{-C}}W^{-p\lambda} + CW^{-\kappa_1/2}.$$

Every path with endpoints in $B_R$ that exits $B_{2W}$ pays this crossing. Thus all paths of cost at most $K$ times the corresponding distance remain in $B_{2W}$ outside the displayed event. The confinement exponent is fixed, with the same permitted exponential deterioration of its prefactor. Markov's inequality now makes the upper references explicit: at failure $\varepsilon$, a variable with the moment bound (7.3) has quantile at most $(C(2+R)^{A}e^{Ca^{-C}}/\varepsilon)^{1/p}$. Together with the fixed inverse moments, this bounds references at failure $e^{-a^{-C}}$ by $e^{Ca^{-C}}$ after enlarging $C$.

*Quantitative choices in the comparison proof.* We record enough of the construction to exclude an ineffective dependence hidden in its constants. The local references and first-exit radii in [2, Section 2] are chosen by inequalities of the form $Hr^{\sigma} < b$. Their inverses are bounded by one exponential of a polynomial in the quantities in (7.3). The annular amplifier [2, Proposition 3.3] uses

$$N = \lceil C(1-q_*)^{-1}\log((C_b+1)/e_c)\rceil, \quad L = \lceil C(1+\beta/\sigma+c_0^{-1})\rceil, \quad a_* = \tfrac{1}{4}L^{-N},$$

where $e_c$ is the coefficient tolerance and $q_* \in (0,1)$ is the contraction factor. Its remaining depth and decay constants are obtained by dividing logarithms of prefactors and inverse error budgets by positive multiples of $\sigma a_*, ha_*, wa_*$, where $h, w > 0$ are the chosen regularity and confinement thresholds. The power $L^N = \exp(N\log L)$ costs one exponentiation, not $N$ nested exponentiations.

The local forcing proof in [2, Section 4] first chooses

$$K_0 > 2\beta + 3, \qquad D_{\mathrm{net}} > 2d(K_0+3)/\sigma + 3, \qquad \mu_0 > D_{\mathrm{net}} + 2.$$

Only then choose the high-confidence reference quantiles. This order is legitimate because site radii $r \in [\Delta^2, \Delta]$ have units between $H^{-1}\Delta^{2\beta}/A_0$ and $H\Delta^{\sigma}/a_{\mathrm{out}}$; increasing $\ell = \log\Delta^{-1}$ absorbs the numerical reference constants without changing $K_0$. The endpoint net has mesh proportional to $(g_\Delta/H)^{1/\sigma}$, so $D_{\mathrm{net}}$ likewise precedes those constants. A local failure $e^{-C(\mu_0+1)}$ gives site-cover failure $\Delta^{\mu_0}$. All these reference quantiles are explicit from (7.3).

Favorable trials are obtained by [2, Lemma 3.4], using Cantelli's inequality and one amplification. If a trial has success at least $p_*$, at most $\lceil Cp_*^{-1}(\mu_0 + \log N_{\mathrm{edges}} + 1)\rceil$ separated trials suffice for the required collector-edge confidence. Their radii, upper and lower primary-unit pin conversions, the mesh clearances in [2, Section 4.1], and the smooth discount in [2, Lemma 4.3] are explicit powers, minima and products of preceding data. The Sobolev derivative order of a discount is fixed by $d$. The resulting catalogue has size at most an exponential of a fixed power of its mesh and arc constants. The primary site reference and its collector are fixed before the seed-trial and pin-conversion quantiles. Those later choices change the number and radii of trials in the reserved boxes; they do not redefine the primary site reference or the collector.

The counting depth is explicit. In the notation of [2, Proposition 4.5, (4.16)–(4.21)], take $\rho = 1/2$. Let $a_{\mathrm{ref}} > 0$ be its saving constant $a_0$, a fixed fraction of the already chosen collar lower bound $a_{\mathrm{out}}$, and let $A_{\mathrm{ref}}$ be its collector-diameter bound $A_0$. Set $\kappa_{\mathrm{count}} = \min\{1/4, a_{\mathrm{ref}}/(16A_{\mathrm{ref}})\}$. Its reciprocal is controlled by those references. Choose

$$k_{\mathrm{count}} \ge \frac{4\mu_0 + 4}{\rho\kappa_{\mathrm{count}}}.$$

Choose $\ell$ at least the maximum of the four quantities

$$\frac{\log(CJ)}{D_0}, \qquad \frac{4[\log k_{\mathrm{count}} + \log(CK_1J)]}{\rho\kappa_{\mathrm{count}}}, \qquad \frac{64k_{\mathrm{count}}D_0M_0^2}{\rho^2\kappa_{\mathrm{count}}^2}, \qquad \rho^{-1}\log k_{\mathrm{count}}.$$

Add the already specified lower bounds for references, unit bins, the endpoint net and prefactors; then choose $\ell$ as twice this finite maximum. Set $\Delta = e^{-\ell}$, and choose the local gain and error as fixed fractions of $vc_{\mathrm{seed}}\Delta^{K_0}/U_0$ and $vc_{\mathrm{seed}}\Delta^{K_0}$, respectively. Choose the drift tolerance as half the minimum of the bounds in the pin cap and [2, (4.25)]. These choices bound reciprocals as well as upper sizes; none is an unspecified positive compactness constant.

The improvement in [2, Propositions 3.5–3.6] uses the same gain $g$, error $\varepsilon_*$ and drift tolerance $t_*$ for every current coefficient in $[1+20v, C_b]$. Its at most $2C_b/g+1$ improvements do not redefine scales or catalogues. One final application of [2, Proposition 3.7] gives the increment estimate. Choose $v$ as a fixed fraction of $\delta$ divided by the fixed guard-diameter quantile.

Here is a finite bound for all these choices. Set

$$B_0 = C(1 + a^{-1} + \delta^{-1})^C, \qquad B_{j+1} = \exp(\exp(B_j^C)).$$

Increase the fixed $C$ before making the choices. A bounded number of products, reciprocals whose positive lower bounds have been recorded, powers $x^y$ with $x, y \le B_j$, and catalogues of size $e^{B_j^C}$ fit under $B_{j+1}$. The preceding construction can be bounded in the following order:

1. $B_1$: regularity moments, inverse exponents, reference quantiles and guard radius;
2. $B_2$: $K_0, D_{\mathrm{net}}, \mu_0$, higher-confidence quantiles and first-exit scales;
3. $B_3$: rough comparison and the amplification producing favorable trials;
4. $B_4$: collector size, trial radii and finite pin conversions;
5. $B_5$: the routing meshes and their clearances;
6. $B_6$: catalogue size, discount amplitude and Cameron–Martin norm;
7. $B_7$: counting multiplicity and the explicit choice of $\ell$;
8. $B_8$: $\Delta^{-1}, g^{-1}, \varepsilon_*^{-1}, t_*^{-1}$;
9. $B_9$: the single amplification used in the fixed-confidence improvement;
10. $B_{10}$: the resulting seed and its parameter-spacing tolerance;
11. $B_{11}$: final amplification depth, prefactor, inverse rate and inverse drift allowance;
12. $B_{12}$: a common upper bound, also including the inverse seed spacing.

Each quantity in a row is obtained from the preceding rows by the displayed choices in the cited proof. In particular, the coefficient iteration does not restart the list. Induction bounds $B_{12}$ by $T_k(C_1(a\delta)^{-C_1})$ for fixed $k, C_1$; additional fixed levels absorb powers of intermediate tower values.

The public comparison thus gives constants $n_0, A, b, t, s > 0$ with $\mathbb{P}[\|F_\zeta - F_\eta\|_S > \delta] \le Ae^{-bn}$ when $n \ge n_0$, $\tau n \le t$ and $\tau \le s$, and bounds $n_0, A, b^{-1}, t^{-1}, s^{-1}$ by a fixed-height tower. Enlarge $\mathcal{R}$ to dominate them. Then $n \ge \mathcal{R}$, $\tau n \le \mathcal{R}^{-1}$ imply $\tau \le \mathcal{R}^{-2} \le s$, and give (7.1). □

# 8 Convergence on fixed surfaces

We first construct the continuous subcritical version needed for the endpoint argument. This also specifies exactly which part of the public continuity proof is used.

**Lemma 8.1.** *After shortening $I = [c_-, c]$ if necessary, the mean-log-normalized metrics have a field-measurable modification on $I \setminus \{c\}$ which is almost surely continuous in the locally uniform topology. On one event, every member of this family is a proper geodesic metric inducing the Euclidean topology.*

*Proof.* Choose a small dyadic $a_*$ below the threshold of Proposition 7.1 for $S = K_1$, and cover $[c - 2a_*, c)$ by the bands $I_a$ with $a = 2^{-j}a_*$. Fix one band. Choose a fixed small $\delta_* > 0$, set $\delta_j = 2^{-j}\delta_*$, and let

$$R_j = \mathcal{R}(a, \delta_j), \quad k_j = \left\lceil T_{k+2}(Aa^{-D}2^{Dj}) \right\rceil, \quad w_j = a2^{-k_j}.$$

Take $A, D$ large enough that $R_j \le T_k(Aa^{-D}2^{Dj})$. At depth $n_j = \lfloor (2R_j w_j)^{-1} \rfloor$, compare every node of the relative dyadic level-$k_{j+1}$ grid with both brackets at level $k_j$. The total failure is at most

$$C2^{k_{j+1}} R_j \exp\{-c_1/(R_j^2 w_j)\}.$$

For $x_j = Aa^{-D}2^{Dj}$, the elementary tower identities give $\log R_j = o(k_j)$ and $\log k_{j+1} = o(k_j)$. Hence the last bound is at most $\exp\{-\exp(c_2 k_j)\}$ for large $j$, and is summable. The depth and drift restrictions hold for those $j$ as well.

Linearly interpolate the $C(K_1^2)$-valued profiles $F_\zeta$ at these grids. Borel–Cantelli makes the sup-norm difference of successive interpolants at most $\delta_j$ eventually. They converge uniformly to a continuous field-measurable profile. For each deterministic $\zeta$, the fixed-accuracy estimate of Proposition 7.1, with depth proportional to $|\zeta - \eta|^{-1}$, gives stochastic continuity as $\eta \to \zeta$. Thus the interpolated limit agrees almost surely with the prescribed profile at that parameter.

To cover space without a radius-dependent parameter cutoff, apply this construction also to $T_R h = h(R\cdot) - h_R(0)$ for integer $R \ge 1$. Each $T_R$ preserves the pinned field law. Exact similarity transports the distance profile on $K_1$ to $K_R$ by the factor $R^{\phi(\zeta)}e^{\zeta h_R(0)}$, which is continuous in $\zeta$. Agreement at countably many dense parameters, including the common band endpoints, patches these profiles on overlapping balls and bands. Exponentiating the interpolated profiles gives a continuous family of finite pseudometrics; the triangle inequality passes from the dense grid parameters.

We now apply the positive-parameter arguments of [2, Sections 5.3–5.4], recording their inputs. Fixed-parameter locality, Weyl scaling and exact similarities are supplied by [1, Definition 1.3 and Theorem 1.4]. On a full-probability event invariant under a countable dense rational subgroup $\mathcal{H}_0$ of bounded pinned smooth Cameron–Martin shifts, continuity extends the fixed-parameter Weyl inequalities to

$$e^{-\zeta\|f\|_\infty}\rho_h^\zeta \le \rho_{h+f}^\zeta \le e^{\zeta\|f\|_\infty}\rho_h^\zeta, \qquad f \in \mathcal{H}_0.$$

For disjoint compact rational balls $A, B$ and a compact parameter band, the set of parameters at which $\rho_h^\zeta(A, B) = 0$ is a random compact set invariant under these shifts. The Gaussian zero-one argument in [2, Section 5.3] makes each of its open-set hitting events deterministic, and hence makes the set deterministic. It is empty because each deterministic marginal is a genuine metric. Countably many ball pairs exclude degeneracy at every parameter on one event.

The dyadic dilation of the sphere-pinned field preserves its law. Intersect the good event with all integer iterates of this dilation; exact covariance holds first at dense parameters and then

everywhere by continuity. On each compact parameter band, the minimum unit-annulus crossing is now positive. The recurrence argument of [2, Section 5.4], using exact similarities, $\inf \phi > 0$ and $h_{2^j}(0) = o(j)$, gives arbitrarily distant annuli whose crossings tend to infinity uniformly on that band. The first-exit inequality passes from dense parameters by profile continuity, so these annuli confine metric balls uniformly. Compactness and separation give the Euclidean topology and properness. Finally, dense-parameter geodesics have compact, equicontinuous subsequences; their arclength equalities pass to the limiting metric and give geodesics for every parameter. These are the arguments of that section before its median-normalization step. All their inputs have just been verified, and no result at the zero-parameter endpoint is used. □

Fix a rational coordinate complex $S$ and work with the modification of Lemma 8.1.

**Proposition 8.2.** *Almost surely,*

$$\lim_{\zeta \uparrow c} \|\rho_h^\zeta - \rho_h^c\|_S = 0,$$

*simultaneously for all rational coordinate complexes $S$.*

*Proof.* By Proposition 5.2, choose a deterministic $\omega(a) \downarrow 0$ such that every deterministic $u \in [c - 2a, c]$ satisfies

$$\mathbb{P}[\|\rho_h^u - \rho_h^c\|_{K_{A_0}} > c_0\omega(a)/4] \le p_0/4.$$

The triangle inequality gives the seed (6.7) for every deterministic pair in this interval at every accuracy $\delta \ge \omega(a)$. Let $k$ be the height in Proposition 7.1, and set

$$L = k + 4, \quad \kappa = \frac{1}{2(M+1)}, \quad \varepsilon_a = \max\{200L\omega(a), a^\kappa\}, \quad \delta_b = \frac{\varepsilon_a}{100L}.$$

For small dyadic $a$, all accuracies are admissible and

$$\delta_b^{M+1}/a \ge c_1 a^{-1/2}. \tag{8.1}$$

Apply Proposition 6.3 to $(u, c)$ at depth $\lfloor c_0\delta_b/(4a) \rfloor$. Both depth restrictions hold by (8.1); hence

$$\sup_{u \in I_a} \mathbb{P}[\|F_u - F_c\|_S > \delta_b] \le C a^{-C} e^{-c_1 a^{-1/2}}. \tag{8.2}$$

The supremum here is outside the probability.

*Fine parameter grids.* Let $G(m, a)$ be the relative dyadic level-$m$ grid in $I_a$, of mesh $a2^{-m}$. Choose fixed $A, D$ sufficiently large and put

$$\delta_j = 2^{-j-10}\varepsilon_a, \quad k_j = \left\lceil T_{k+2}(Aa^{-D}2^{Dj}) \right\rceil, \quad w_j = a2^{-k_j}, \quad R_j = \mathcal{R}(a, \delta_j).$$

Compare each node of $G(k_{j+1}, a)$ with both its brackets in $G(k_j, a)$, using depth $n_j = \lfloor (2R_j w_j)^{-1} \rfloor$. Proposition 7.1 gives total level-$j$ failure at most

$$C 2^{k_{j+1}} R_j \exp\{-c_1/(R_j^2 w_j)\}. \tag{8.3}$$

Here are the growth checks. Since $\varepsilon_a \ge a^\kappa$, choose $D > C_1(1 + \kappa) + C_1 + 2$ so $R_j \le T_k(x_j)$ with $x_j = Aa^{-D}2^{Dj}$. Then $\log R_j = o(k_j)$ uniformly in $j$ as $a \downarrow 0$. For fixed $m \ge 1, B > 1$, $\log T_m(Bx) = o(T_m(x))$; induction on $m$ proves this directly. Hence $\log k_{j+1} = o(k_j)$ uniformly, and eventually

$$\frac{1}{R_j^2 w_j} \ge e^{(\log 2)k_j/2}, \qquad k_{j+1} + \log R_j \le e^{(\log 2)k_j/4}.$$

These bounds verify $n_j \geq R_j$, $w_j n_j \leq R_j^{-1}$ and bound (8.3) by $\exp\{-\exp(c_1 k_j)\}$. For $a = 2^{-m}$, grouping by $m + j$ proves joint summability.

Let $\mathcal{F}_j$ be the piecewise linear $C(S^2)$-valued interpolant on $G(k_j, a)$. Successful bracket comparisons give $\sup_{u \in I_a} \|\mathcal{F}_{j+1}(u) - \mathcal{F}_j(u)\|_S \leq \delta_j$: the inequality holds at each new node, and convexity gives it between nodes. The already continuous parameter trajectory is the uniform limit of these interpolants. Thus

$$\sup_{u \in I_a} \|F_u - \mathcal{F}_0(u)\|_S \leq \sum_{j \geq 0} \delta_j < \varepsilon_a/500 \tag{8.4}$$

whenever all fine comparisons succeed.

*Reaching the first fine grid.* Choose $A_1 > 2D + 4$ and $\ell_0 = \lceil A_1 \log_2(a^{-1}) \rceil$. This grid has polynomial cardinality, so (8.2) controls all its nodes with summable failure. Starting from $\ell_0$, set

$$\ell_{i+1} = \min\{k_0, \lfloor 2^{\ell_i/2} \rfloor\}$$

until $k_0$ is reached. There are at most $L$ steps for small $a$. Indeed, with $x = Aa^{-D}$, the first new level is at least $8x$. For large $u$, $\lfloor 2^{u/2} \rfloor \geq e^{u/4}$, and $u \geq 8T_r(x)$ implies $e^{u/4} \geq 8T_{r+1}(x)$. Induction reaches $T_{k+2}(x)$ within $k + 3$ steps, with the extra step absorbing rounding.

At each bridge compare every new node with one old bracket. Its spacing is at most $w = a2^{-\ell_i}$. Use the seeded estimate at accuracy $\delta_b$ and depth $\lfloor c_0 \delta_b/(2w) \rfloor$. The total failure is at most

$$C 2^{\ell_{i+1}} \delta_b^{-M} \exp\{-c_1 \delta_b^{M+1} a^{-1} 2^{\ell_i}\} \leq \exp\{-c_1 a^{-1/2} 2^{\ell_i}\},$$

by (8.1) and $\ell_{i+1} \leq 2^{\ell_i/2}$. These failures are summable over dyadic bands and the at most $L$ bridges. The accumulated error is at most $L\delta_b = \varepsilon_a/100$.

Borel–Cantelli now applies jointly to the coarse, bridge and fine failures. Almost surely, all comparisons succeed on all sufficiently small dyadic bands, and (8.4) gives $\sup_{u \in I_a} \|F_u - F_c\|_S \leq \varepsilon_a$. The bands cover a punctured left neighborhood of $c$, and $\varepsilon_a \to 0$. The limiting critical profile is bounded, so exponentiation converts logarithmic-profile convergence to distance convergence. Finally intersect over the countable collection of rational complexes. □

## 9 A common spatial modulus

Surface convergence lets us choose the fine-field percolation event for the whole parameter family at once.

**Lemma 9.1.** *Almost surely, for each rational closed cube $C$ and open neighborhood $U$ of its boundary,*

$$\sup_{\zeta \in I} \operatorname{diam}_{\rho_h^\zeta(\cdot,\cdot;U)} \partial C < \infty.$$

*Proof.* Choose a bounded rational polycubical $V$ with $\partial C \subset V \Subset U$, and a rational coordinate complex containing $\partial C \cup \partial V$ and connecting rectangles. Proposition 8.2 and interior parameter continuity give joint continuity on $I$ times the square of that complex. Every member is a genuine metric: this holds simultaneously below $c$ by Lemma 8.1, and at $c$ by Theorem A. Therefore

$$b = \min_{\zeta \in I,\ x \in \partial C,\ y \in \partial V} \rho_h^\zeta(x, y) > 0.$$

Joint uniform continuity supplies $\delta > 0$ such that every $\delta$-close boundary pair has distance below $b/2$ for every parameter. Near-minimizing paths cannot leave $V$, so global and intrinsic distances

agree on these pairs. A uniformly bounded number of $\delta$-small steps along cube faces connects any boundary pair. This proves the bound, and also spatial continuity of each intrinsic boundary restriction. □

Fix a countable dense $\Lambda \subset I$ containing $c$. Intersect the needed Weyl and similarity events for its parameters and the countably many templates and scales. From $h = X + f$ and the local Weyl identity, Lemma 9.1 gives a finite random bound for $\sup_{\zeta\in\Lambda} \operatorname{diam}_{\rho_X^\zeta(\cdot,\cdot;U_0)} \partial C_0$. Choose $A$ so its upper-tail probability is below the fixed percolation threshold in Proposition 3.2. At mesh $N^{-1}$, declare a tile good if

$$\operatorname{diam}_{\rho^\zeta_{X-H_{\log N}}(\cdot,\cdot;\text{tile buffer})} \partial C_v \le AN^{-\phi(\zeta)} \quad \text{for every } \zeta \in \Lambda.$$

This is one measurable local fine-field event because $\Lambda$ is countable. Exact similarities give the same failure bound at every tile. The labels have a fixed finite dependence range.

The lattice construction of Proposition 3.2 now uses the same good-tile configuration for every parameter. The paths may depend on the parameter. If $Y^\zeta$ is the corresponding unit separator-plus-crossing functional, that proof gives

$$Y^* := \sup_{\zeta\in\Lambda} Y^\zeta \le 1 + CAN^{d-\inf_I \phi} \exp\left(c \sup_{B_{200}} H^+_{\log N}\right)$$

outside probability $CN^d e^{-c_1 N}$. Taking $N = \lceil u^2 \rceil$ gives

$$\mathbb{P}[Y^* > e^u] \le C \exp(-c_1 u^2/\log u). \tag{9.1}$$

This is a joint percolation estimate, not a union bound over parameters.

At depth $n$, the rescaled fine-field $Y^*_{n,z}$ is independent of the entire coarse noise. Set $Z^*_n(z) = (1+c)R_n(z) + \log(4Y^*_{n,z})$. The conditional estimate (3.4) and Lemma 3.3 hold for $Z^*_n$, by (9.1). Equation (5.13) then bounds every separator and crossing cost, for all $\zeta \in \Lambda$, by $e^{-\eta V_n(x)}$ eventually, for each fixed $\eta < c_-$. Corollary 4.3 and the same finite-chain gluing as in Theorem A give, almost surely,

$$\sup_{\zeta\in\Lambda} \rho_h^\zeta(x,y) \le C_{K,\theta} \left(\log \frac{1}{|x-y|}\right)^{-\theta}, \qquad 0 < \theta < \frac{c_-}{2q}, \tag{9.2}$$

for all $x, y \in K$ with $0 < |x-y| \le 1/2$. All paths remain in one compact enlargement, so the smooth correction is bounded uniformly by $e^{c\sup|f|}$. Interior parameter continuity extends (9.2) from $\Lambda$ to every $\zeta < c$; $c$ was included from the start.

Choose a finite rational $r$-net $\{p_i\}$ for $K_R$, with points in $K_{R+1}$, and include it in a rational coordinate complex. Surface convergence gives $\max_{i,j} |\rho_h^\zeta(p_i,p_j) - \rho_h^c(p_i,p_j)| \to 0$ almost surely. The triangle inequality and (9.2) give

$$\|\rho_h^\zeta - \rho_h^c\|_{K_R} \le \max_{i,j} |\rho_h^\zeta(p_i,p_j) - \rho_h^c(p_i,p_j)| + 4C_{R+1,\theta}(\log(1/r))^{-\theta}.$$

First send $\zeta \uparrow c$, then $r \downarrow 0$ along a deterministic sequence. Intersect over integer $R$. This proves the locally uniform almost-sure endpoint limit in mean-log normalization. The conclusions concern only $h$ and its metric family, so the use of an enlarged white-noise coupling does not change their probability under the prescribed field law.

## 10 Median normalization

We finish by checking the deterministic normalization constants.

**Lemma 10.1.** *For every deterministic $\zeta \in I$, the law of $\rho_h^\zeta(0, e_1)$ has full support in $(0, \infty)$. Its lower median is positive, finite and unique.*

*Proof.* A compactly supported smooth function of zero unit-sphere average is a Cameron–Martin perturbation of the pinned field: its squared norm is a positive constant times $\int |\xi|^d |\widehat{f}(\xi)|^2 \, d\xi < \infty$. Choose a bounded connected tube $U$ around $[0, e_1]$ whose closure misses a sphere patch near $-e_1$. A compactly supported smooth $f$ can equal $-1$ on $U$ and have zero sphere average, by correcting the average on that disjoint patch. Internal distance in $U$ is finite. Thus

$$\rho_{h+tf}^\zeta(0, e_1) \le e^{-\zeta t} \rho_h^\zeta(0, e_1; U) \longrightarrow 0.$$

For every $a > 0$, a countable union over integer $t$ and Cameron–Martin equivalence give

$$\mathbb{P}[\rho_h^\zeta(0, e_1) < a] > 0.$$

Next choose smooth $u \ge 0$, supported in the Euclidean ball of radius $1/2$ and equal to one on the Euclidean annulus with radii $1/8, 1/4$. It has zero unit-sphere average. Set $D_t = \rho_{h+tu}^\zeta$. For $t > s$, every path from zero to $e_1$ crosses this annulus, so the Weyl length identity gives

$$D_t(0, e_1) \ge D_s(0, e_1) + (e^{\zeta(t-s)} - 1) D_s(\partial B_{1/8}^{\mathrm{Eucl}}, \partial B_{1/4}^{\mathrm{Eucl}}).$$

The last distance is positive by continuity and separation on disjoint compact boundaries. Bounded Weyl scaling makes $t \mapsto D_t(0, e_1)$ continuous; the displayed inequality makes it strictly increasing and unbounded. On the event of distance below $a$, some nonnegative rational $t$ puts the distance in any prescribed interval $(a, b)$. Countability and Cameron–Martin equivalence give positive probability to that interval. Full support makes the distribution function strictly increasing. The lower median is therefore unique; atomlessness is not needed. □

*Completion of Theorem B.* Write $m(\zeta) = \operatorname{med} \rho_h^\zeta(0, e_1)$. The established endpoint limit implies convergence in distribution of these unit distances. By Lemma 10.1, continuity points on either side of $m(c)$ can be chosen arbitrarily close to it, with distribution values strictly below and above $1/2$. Weak convergence traps the nearby medians between those points. Hence $m(\zeta) \to m(c) \in (0, \infty)$. Since these numbers are deterministic,

$$\| m(\zeta)^{-1} \rho_h^\zeta - m(c)^{-1} \rho_h^c \|_{K_R} \longrightarrow 0 \quad \text{almost surely.}$$

This is (1.3) in the prescribed median normalization. For any $\theta < c/(2q)$, choose the initial interval short enough that $c_- > 2q\theta$. Equation (9.2) and the bounded reciprocal medians near $c$ then prove (1.4). □

## References

[1] *Supercritical exponential metrics in higher dimensions: Tightness, existence, and uniqueness.* Research manuscript, public 94-page version, 26 October 2026. Public manuscript.

[2] *Almost sure continuity of the higher-dimensional LQG metric.* Research manuscript, public 34-page version. Public manuscript.

[3] J. Ding and E. Gwynne. *The critical Liouville quantum gravity metric induces the Euclidean topology.* arXiv:2108.12067.

[4] J. Ding, R. Roy and O. Zeitouni. *Convergence of the centered maximum of log-correlated Gaussian fields.* arXiv:1503.04588, Proposition 1.1.

[5] Á. Timár. *Boundary-connectivity via graph theory.* arXiv:0711.1713, Theorem 4.

[6] J. Ding, E. Gwynne and Z. Zhuang. *Tightness of exponential metrics for log-correlated Gaussian fields in arbitrary dimension.* arXiv:2310.03996, version 5.